# 🧪 Digital Forensics Case Study

## Insider Intellectual Property Theft – ShockWave Analytics

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## 📘 Case Background

ShockWave Analytics is a mid-sized cybersecurity software company specialising in predictive threat detection tools for financial institutions. The company has been developing a new proprietary machine-learning engine:*Project Triton*, expected to generate significant competitive advantage.

John Matthews, a senior software engineer, had been employed at ShockWave for five years. He was one of four developers with full access to the Triton source code repository and design documentation.

### Timeline of Events

- **March 3** – John submits formal resignation, stating he accepted a position at a "different industry startup."
- **March 17** – John's final working day.
- **March 28** – A competitor, BasilCore Systems, announces a new product with architecture remarkably similar to Triton.
- **April 2** – ShockWave's internal review discovers unusual file access patterns in the two weeks before John's departure.

Senior management suspects potential intellectual property theft and contacts the legal department.
A formal digital forensic investigation is initiated to determine:

- Whether proprietary information was stolen
- Whether John acted alone or in collaboration
- Whether the company has grounds for civil litigation

ShockWave IT preserved:

- *John's company-issued laptop, A USB storage device found in his desk drawer, Relevant network logs, Email server archives, File server access logs, HR resignation and termination documentation*

You are assigned as forensic analysts engaged by ShockWave's legal team.
**The task is to conduct a forensic analysis**

# Evidence Sources Available

Forensic image of John's company laptop

Forensic image of USB storage device recovered from his desk

Corporate firewall logs

Email server logs and mailbox archive

Windows Security and System Event Logs from the laptop

File server access logs (Project Triton directory)

HR resignation and exit timeline documentation

# Evidence-by-Evidence Breakdown

### 1. Laptop Image

Possible findings:

- Encrypted archive
- Prefetch files
- Registry USB history
- Browser search history
- Email drafts
- Wiping software traces
- $MFT timestamp analysis

### 2. USB Image

Possible findings:

- Triton_backup.7z
- Volume serial number
- File creation timestamps
- Partial deleted files

### 3. Firewall Logs

Possible findings:

- Outbound SMTP traffic
- Gmail HTTPS connections
- Large data transfer sessions

### 4. Email Logs

Possible findings:

- Sent message with encrypted attachment
- Message-ID and SMTP routing
- Timestamp correlation

### 5. Windows Event Logs

Possible findings:

- USB device connection events
- Logon events
- After-hours activity
- Application execution logs

### 6. File Server Logs

Possible findings:

- Bulk read/download pattern
- Access outside normal working hours
- No similar activity from other employees

### 7. HR Documentation

Possible Findings

- Employment timeline
- Motive window
- Access privileges
- Exit interview statements (e.g., "no future plans yet")

# Timeline of John's Malicious Activity

## Phase 1 – Pre-Incident Indicators (Motive Development)

March 1 – 18:42
Browser artifact: LinkedIn messages with BasilCore recruiter.
Google searches:

- "Senior ML salary comparison"
- "non compete enforceability"

March 3 – 09:15
*HR record*: John submits resignation letter.
States: "Pursuing opportunity in different industry."
Interpretation: ?

## Phase 2 – Reconnaissance & Preparation

March 8 - 20:11
*File Server Log:*
Access to \\\\BlueWave-FS01\\Projects\\Triton\\
Bulk directory listing performed.

20:18
Windows Security Log:

- Logon type 2 (interactive)
- After-hours login

20:21
Prefetch created:
7ZFM.EXE-3FA92BC1.pf
First recorded execution.
Interpretation: ?

# Phase 3 – Data Collection & Staging

## March 10 (Primary Theft Night)

### 1 21:43:12 – USB Device Inserted

Event Log ID 2003

Registry:
HKLM\\SYSTEM\\CurrentControlSet\\Enum\\USBSTOR\\

Device: SanDisk Ultra 128GB
Serial: 4C530001230219117381

### 2 21:45 – 22:12 – Bulk File Access

File Server Logs show:

- 4,283 file read operations
- 14.2 GB transferred

Mostly: /SourceCode/, /ModelWeights/, /DesignDocs/

### 3 22:14 – Local Directory Created

C:\\Users\\John\\AppData\\Local\\Temp\\Triton_Backup\\

- $MFT shows creation timestamp.
- Last accessed: 22:15.

# Phase 4 – Archive Creation

## March 11

### 00:14:33 – Encrypted Archive Created

**Triton_full_backup.7z**

#### Prefetch

7Z.EXE-3A4F2B1C.pf

- Run count: 2
- Last run: 00:14

#### $MFT

- Created: 00:14:33
- Size: 13.8 GB

#### Registry

RecentDocs entry for Triton_full_backup.7z

🔎 Interpretation?

# Phase 5 – Data Exfiltration

### 🕐 00:18:02 – File Copied to USB

- $LogFile artifacts show file write to removable volume.
- Volume serial matches USB registry entry.

### 🕐 00:22:47 – Email Sent to Personal Gmail

**Email server logs:**

- Sender: j.matthews@bluewave.com
- Recipient: john.private@gmail.com
- Attachment: Triton_full_backup.7z (password protected)

## Supporting Artifacts

### Firewall Logs

Outbound SMTP session; 13.8 GB encrypted HTTPS traffic to Gmail servers.

### Browser History

Gmail login at 00:21:10; Search: "how to email large encrypted files"

**Interpretation:?**

# Phase 6 – Anti-Forensics

## March 12

### 🕐 00:16:09 – Prefetch

ERASER.EXE-91BC22AD.pf

- Run count: 3
- Last run: 00:16

### 🕐 00:17 – 00:20 – Windows Event Log

- Application execution logs
- File deletion entries

### Unallocated Space

- Partial remnants of: Triton_Backup temp directory

### Browser Search History

> "best file shredder windows free"
>
> "does deleting prefetch remove evidence"

**Interpretation?**

# Phase 7 – Post-Incident Behaviour

### March 15

- USB device reinserted (Event Log ID 2003).
- No major file transfer.
- Possibly verifying contents.

### March 17

- Final working day.
- Account disabled at 18:03 (IT log).

### March 28

- BasilCore announces similar ML engine.
- ShockWave initiates internal review.